

Prompt Engineering — Quiz 6

Name: _____ Date: _____

Instructions: This quiz contains 12 multiple-choice questions based on the "Ethics, Copyright, and Responsible Use" (Week 12) and "Emerging Trends" (Week 13) lectures. Answer all questions.

#1 What phenomenon occurs when new AI models are trained on data generated by previous AI models, potentially amplifying biases and reducing quality?

- A) Transfer Learning
- B) Model Collapse (Recursive Contamination)
- C) Zero-Shot Learning
- D) Reinforcement Learning

#2 Under current US Copyright Office guidance, which of the following is true regarding AI-generated content?

- A) All AI-generated content is automatically copyrighted.
- B) Works created solely by a machine without human intervention are not eligible for copyright.
- C) The AI model itself owns the copyright.
- D) Prompt engineering alone is always sufficient to claim full authorship.

#3 What is "Prompt Injection"?

- A) Adding more context to a prompt to make it clearer.
- B) A security vulnerability where malicious inputs manipulate the model to bypass safety filters.
- C) Injecting data into the training set.
- D) Using a prompt to speed up the model's response time.

#4 Which of the following practices is banned under the "Unacceptable Risk" category of the EU AI Act?

- A) Using chatbots for customer service.
- B) Social scoring systems that classify people based on behavior or status.
- C) Generative AI for creative writing.
- D) Automated email filtering.

#5 In the context of "Responsible Prompt Design," what is the purpose of a "Refusal Mechanism"?

- A) To make the model refuse to answer any question.
- B) To instruct the model to politely decline sensitive, harmful, or out-of-scope requests.
- C) To prevent the user from accessing the model.
- D) To stop the model from hallucinating.

#6 How does an **AI Agent** primarily differ from a standard **AI Assistant** (Chatbot)?

- A) Agents are slower than assistants.
 - B) Agents are proactive, capable of multi-step planning, and can autonomously use tools.
 - C) Assistants can only understand text, while agents only understand images.
 - D) There is no difference; the terms are interchangeable.
- #7** Which advanced reasoning architecture models thought processes as a network, allowing for non-linear reasoning and combining disparate information paths?
- A) Chain-of-Thought (CoT)
 - B) Tree of Thoughts (ToT)
 - C) Graph of Thoughts (GoT)
 - D) Skeleton-of-Thought (SoT)
- #8** What is the primary advantage of the **Skeleton-of-Thought (SoT)** technique?
- A) It increases the creativity of the model.
 - B) It reduces latency by generating an outline first and expanding points in parallel.
 - C) It ensures the model never hallucinates.
 - D) It allows the model to generate images.
- #9** What problem does the **DSPy** framework aim to solve?
- A) The high cost of API calls.
 - B) The brittleness of hand-tuned prompts by abstracting them into code and optimizing them automatically.
 - C) The lack of multimodal capabilities in older models.
 - D) The difficulty of installing Python packages.
- #10** What is the function of **Temporal** in an agentic workflow?
- A) It generates the text for the agent.
 - B) It provides a stateful workflow engine to ensure durable execution and zero data loss.
 - C) It is a vector database for memory.
 - D) It is a tool for generating images.
- #11** Which defense strategy involves wrapping user input in rigid structures (e.g., "Evaluate this request for safety before executing")?
- A) Prompt Injection
 - B) Prompt Scaffolding
 - C) Sandboxing
 - D) Fine-tuning
- #12** What is **Chain of Draft (CoD)** designed to optimize?
- A) Image generation quality.
 - B) Token usage and latency by generating minimalistic, information-dense reasoning.
 - C) The emotional tone of the response.
 - D) The diversity of the training data.